

```

%% =====
% Frequency Division Multiplexing (FDM) using MATLAB
% Author : Suraj Jha
% Description:
% This program demonstrates Frequency Division Multiplexing
% and Demultiplexing using AM modulation and filtering.
%% =====

clc;
clear;
close all;

%% =====
% MESSAGE SIGNAL GENERATION
%% =====

fs = 4001; % Sampling Frequency
t = 0:1/fs:1-1/fs; % Time vector

% Message Signal 1
Am1 = 2; % Amplitude
fm1 = 4; % Frequency
m1 = Am1*cos(2*pi*fm1*t);

% Message Signal 2
Am2 = 3;
fm2 = 5;
m2 = Am2*cos(2*pi*fm2*t);

% Message Signal 3
Am3 = 4;
fm3 = 6;
m3 = Am3*cos(2*pi*fm3*t);

%% =====
% CARRIER SIGNAL GENERATION
%% =====

% Carrier Signal 1
Cm1 = 1;
fc1 = 100;
c1 = Cm1*cos(2*pi*fc1*t);

% Carrier Signal 2
Cm2 = 1;
fc2 = 170;
c2 = Cm2*cos(2*pi*fc2*t);

% Carrier Signal 3
Cm3 = 1;

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fc3 = 250;
c3 = Cm3*cos(2*pi*fc3*t);

%% =====
%                               COMPOSITE FDM SIGNAL GENERATION
%% =====

x = (m1 .* c1) + (m2 .* c2) + (m3 .* c3);

%% =====
%                               MESSAGE SIGNALS (TIME DOMAIN)
%% =====

figure;

subplot(3,1,1)
plot(t,m1)
xlabel('Time (s)')
ylabel('Amplitude')%%
=====
%   Frequency Division Multiplexing (FDM) using MATLAB
%   Author : Suraj Jha
%   Description:
%   This program demonstrates Frequency Division Multiplexing
%   and Demultiplexing using AM modulation and filtering.
%% =====

clc;
clear;
close all;

%% =====
%                               MESSAGE SIGNAL GENERATION
%% =====

fs = 4001;
t = 0:1/fs:1-1/fs;

Am1 = 2;
fm1 = 4;
m1 = Am1*cos(2*pi*fm1*t);

Am2 = 3;
fm2 = 5;
m2 = Am2*cos(2*pi*fm2*t);

Am3 = 4;
fm3 = 6;
m3 = Am3*cos(2*pi*fm3*t);

```

```

%% =====
%                               CARRIER SIGNAL GENERATION
%% =====

Cm1 = 1;
fc1 = 100;
c1 = Cm1*cos(2*pi*fc1*t);

Cm2 = 1;
fc2 = 170;
c2 = Cm2*cos(2*pi*fc2*t);

Cm3 = 1;
fc3 = 250;
c3 = Cm3*cos(2*pi*fc3*t);

%% =====
%                               COMPOSITE FDM SIGNAL GENERATION
%% =====

x = (m1 .* c1) + (m2 .* c2) + (m3 .* c3);

%% =====
%                               MESSAGE SIGNALS (TIME DOMAIN)
%% =====

figure;

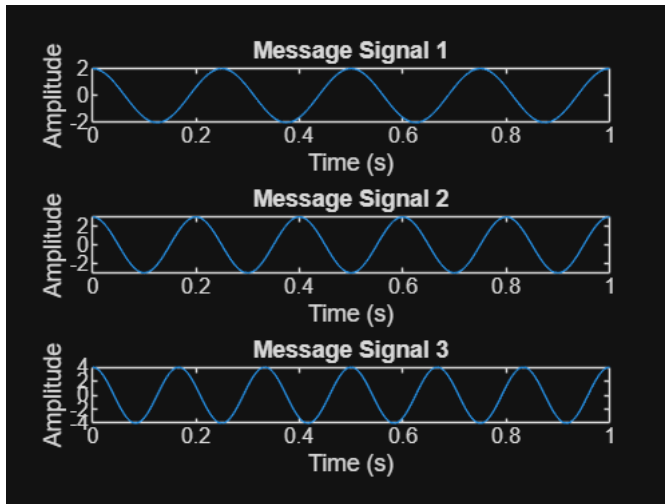
subplot(3,1,1)
plot(t,m1)
xlabel('Time (s)')
ylabel('Amplitude')
title('Message Signal 1')
ylim([-Am1 Am1])

subplot(3,1,2)
plot(t,m2)
xlabel('Time (s)')
ylabel('Amplitude')
title('Message Signal 2')
ylim([-Am2 Am2])

subplot(3,1,3)
plot(t,m3)
xlabel('Time (s)')
ylabel('Amplitude')
title('Message Signal 3')
ylim([-Am3 Am3])

exportgraphics(gcf, 'Figure1_MessageSignals.png', 'Resolution', 300);

```



```
%% =====
%                               FREQUENCY DOMAIN ANALYSIS
%% =====

M1 = abs(fftshift(fft(m1)))/(fs/2);
M2 = abs(fftshift(fft(m2)))/(fs/2);
M3 = abs(fftshift(fft(m3)))/(fs/2);

X = abs(fftshift(fft(x)))/(fs/2);

f = fs/2 * linspace(-1,1,fs);

%% =====
%                               MESSAGE SIGNALS (FREQUENCY DOMAIN)
%% =====

figure;

subplot(3,1,1)
stem(f,M1)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Message Signal 1 Spectrum')
axis([-10 10 0 2.5])

subplot(3,1,2)
stem(f,M2)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Message Signal 2 Spectrum')
axis([-10 10 0 3.5])

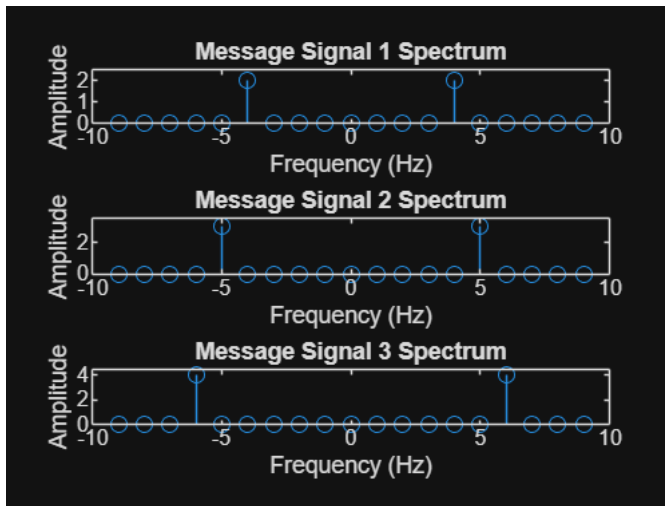
subplot(3,1,3)
stem(f,M3)
```

```

xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Message Signal 3 Spectrum')
axis([-10 10 0 4.5])

exportgraphics(gcf, 'Figure2_MessageSpectrum.png', 'Resolution', 300);

```



```

%% =====
%                               CARRIER SIGNALS (TIME DOMAIN)
%% =====

figure;

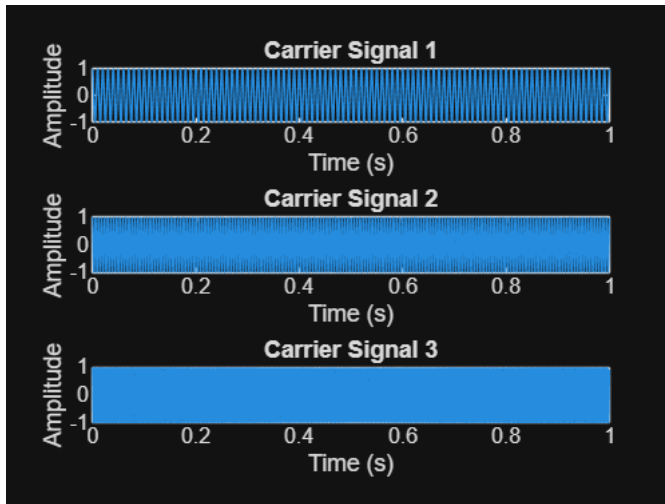
subplot(3,1,1)
plot(t,c1)
xlabel('Time (s)')
ylabel('Amplitude')
title('Carrier Signal 1')
ylim([-Cm1 Cm1])

subplot(3,1,2)
plot(t,c2)
xlabel('Time (s)')
ylabel('Amplitude')
title('Carrier Signal 2')
ylim([-Cm2 Cm2])

subplot(3,1,3)
plot(t,c3)
xlabel('Time (s)')
ylabel('Amplitude')
title('Carrier Signal 3')
ylim([-Cm3 Cm3])

exportgraphics(gcf, 'Figure3_CarrierSignals.png', 'Resolution', 300);

```



```
%% =====
%           CARRIER SIGNALS (FREQUENCY DOMAIN)
%% =====

C1 = abs(fftshift(fft(c1)))/(fs/2);
C2 = abs(fftshift(fft(c2)))/(fs/2);
C3 = abs(fftshift(fft(c3)))/(fs/2);

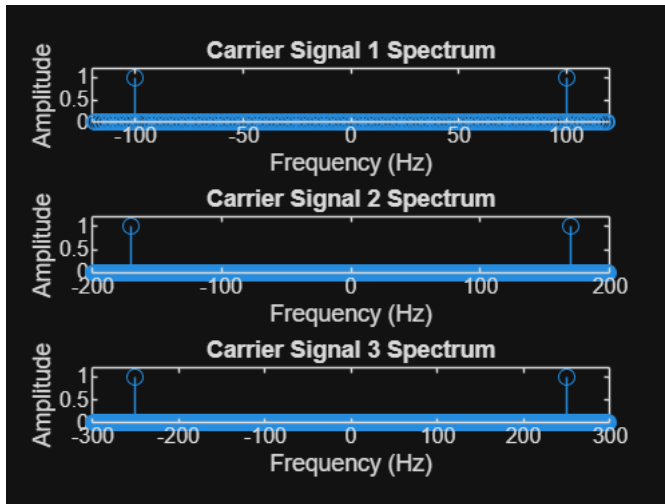
figure;

subplot(3,1,1)
stem(f,C1)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Carrier Signal 1 Spectrum')
axis([-120 120 0 1.2])

subplot(3,1,2)
stem(f,C2)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Carrier Signal 2 Spectrum')
axis([-200 200 0 1.2])

subplot(3,1,3)
stem(f,C3)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Carrier Signal 3 Spectrum')
axis([-300 300 0 1.2])

exportgraphics(gcf, 'Figure4_CarrierSpectrum.png', 'Resolution', 300);
```



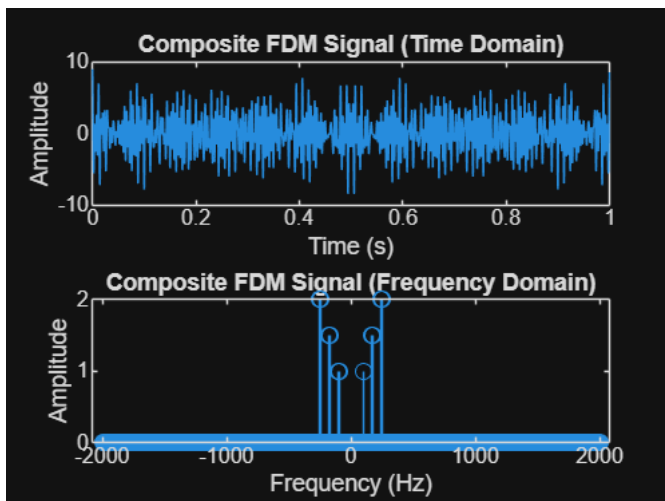
```
%% =====
%                               COMPOSITE FDM SIGNAL
%% =====

figure;

subplot(2,1,1)
plot(t,x)
xlabel('Time (s)')
ylabel('Amplitude')
title('Composite FDM Signal (Time Domain)')

subplot(2,1,2)
stem(f,X)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Composite FDM Signal (Frequency Domain)')

exportgraphics(gcf, 'Figure5_CompositeFDM.png', 'Resolution', 300);
```



```

%% =====
%           BANDPASS FILTERING
%% =====

[num1, den1] = butter(5,[(fc1-fm1-6)/(fs/2),(fc1+fm1+6)/(fs/2)]);
bpf1 = filter(num1,den1,x);

[num2, den2] = butter(5,[(fc2-fm2-6)/(fs/2),(fc2+fm2+6)/(fs/2)]);
bpf2 = filter(num2,den2,x);

[num3, den3] = butter(5,[(fc3-fm3-6)/(fs/2),(fc3+fm3+6)/(fs/2)]);
bpf3 = filter(num3,den3,x);

%% =====
%           DEMODULATION
%% =====

z1 = 2*bpf1.*c1;
z2 = 2*bpf2.*c2;
z3 = 2*bpf3.*c3;

%% =====
%           LOW PASS FILTERING
%% =====

[num4,den4] = butter(5,(fm1+3)/(fs/2));
rec1 = filter(num4,den4,z1);

[num5,den5] = butter(5,(fm2+3)/(fs/2));
rec2 = filter(num5,den5,z2);

[num6,den6] = butter(5,(fm3+3)/(fs/2));
rec3 = filter(num6,den6,z3);

%% =====
%           RECOVERED SIGNALS
%% =====

figure;

subplot(3,1,1)
plot(t,rec1)
xlabel('Time (s)')
ylabel('Amplitude')
title('Recovered Signal 1')
ylim([-Am1 Am1])

subplot(3,1,2)
plot(t,rec2)

```



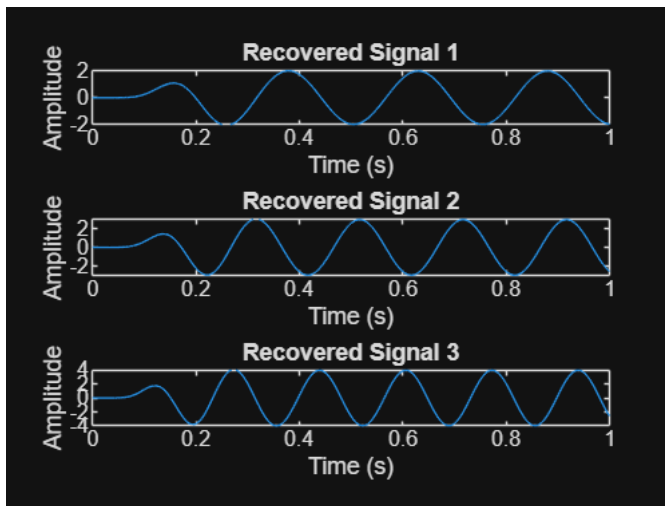
```

xlabel('Time (s)')
ylabel('Amplitude')
title('Recovered Signal 2')
ylim([-Am2 Am2])

subplot(3,1,3)
plot(t,rec3)
xlabel('Time (s)')
ylabel('Amplitude')
title('Recovered Signal 3')
ylim([-Am3 Am3])

exportgraphics(gcf,'Figure6_RecoveredSignals.png','Resolution',300);

```



```

%% =====
%               RECOVERED SPECTRUM
%% =====

R1 = abs(fftshift(fft(rec1)))/(fs/2);
R2 = abs(fftshift(fft(rec2)))/(fs/2);
R3 = abs(fftshift(fft(rec3)))/(fs/2);

figure;

subplot(3,1,1)
stem(f,R1)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Recovered Signal 1 Spectrum')
xlim([-10 10])

subplot(3,1,2)
stem(f,R2)
xlabel('Frequency (Hz)')
ylabel('Amplitude')

```

```

title('Recovered Signal 2 Spectrum')
xlim([-10 10])

subplot(3,1,3)
stem(f,R3)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Recovered Signal 3 Spectrum')
xlim([-10 10])

exportgraphics(gcf,'Figure7_RecoveredSpectrum.png','Resolution',300);

%% =====
%                               END OF PROGRAM
%% =====

disp('All figures have been exported successfully!');

```

All figures have been exported successfully!

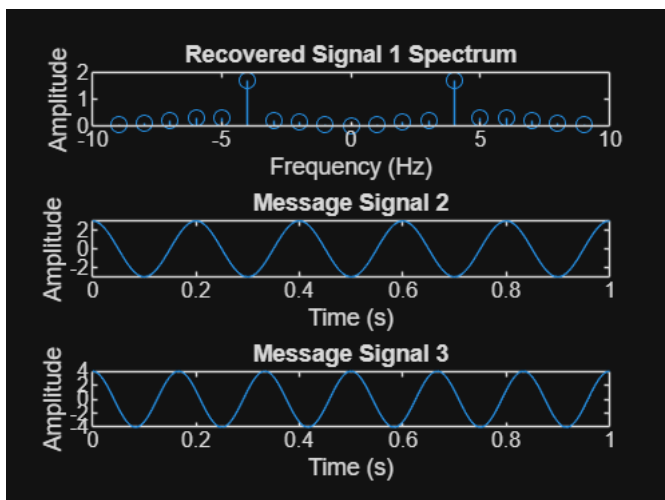
```

title('Message Signal 1')
ylim([-Am1 Am1])

subplot(3,1,2)
plot(t,m2)
xlabel('Time (s)')
ylabel('Amplitude')
title('Message Signal 2')
ylim([-Am2 Am2])

subplot(3,1,3)
plot(t,m3)
xlabel('Time (s)')
ylabel('Amplitude')
title('Message Signal 3')
ylim([-Am3 Am3])

```



```

%% =====
%           FREQUENCY DOMAIN ANALYSIS
%% =====

M1 = abs(fftshift(fft(m1)))/(fs/2);
M2 = abs(fftshift(fft(m2)))/(fs/2);
M3 = abs(fftshift(fft(m3)))/(fs/2);

X = abs(fftshift(fft(x)))/(fs/2);

f = fs/2 * linspace(-1,1,fs);

%% =====
%           MESSAGE SIGNALS (FREQUENCY DOMAIN)
%% =====

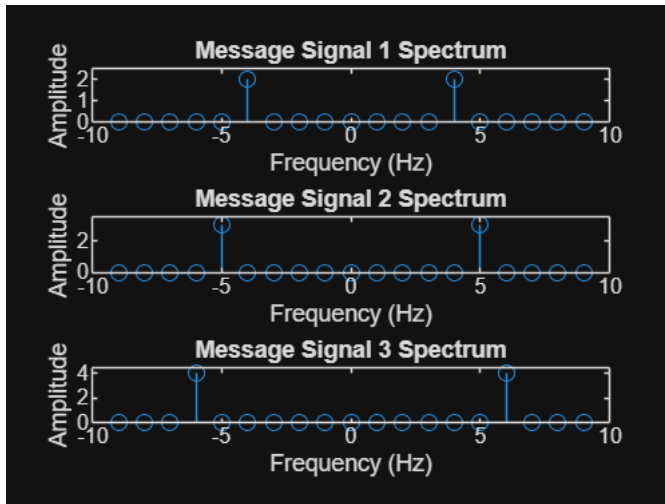
figure;

subplot(3,1,1)
stem(f,M1)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Message Signal 1 Spectrum')
axis([-10 10 0 2.5])

subplot(3,1,2)
stem(f,M2)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Message Signal 2 Spectrum')
axis([-10 10 0 3.5])

subplot(3,1,3)
stem(f,M3)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Message Signal 3 Spectrum')
axis([-10 10 0 4.5])

```



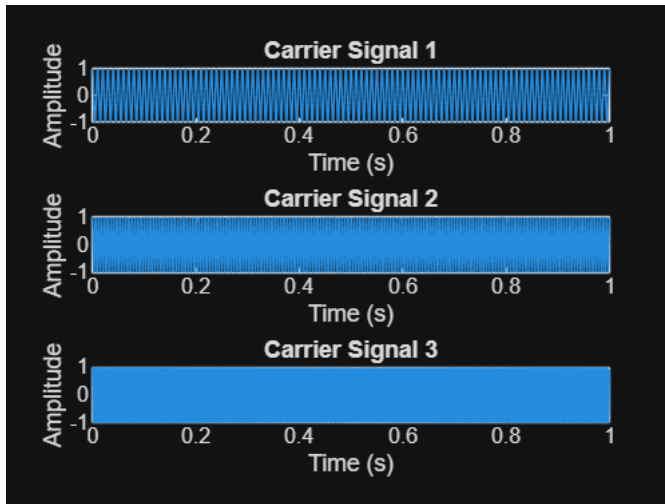
```
%% =====
%           CARRIER SIGNALS (TIME DOMAIN)
%% =====

figure;

subplot(3,1,1)
plot(t,c1)
xlabel('Time (s)')
ylabel('Amplitude')
title('Carrier Signal 1')
ylim([-Cm1 Cm1])

subplot(3,1,2)
plot(t,c2)
xlabel('Time (s)')
ylabel('Amplitude')
title('Carrier Signal 2')
ylim([-Cm2 Cm2])

subplot(3,1,3)
plot(t,c3)
xlabel('Time (s)')
ylabel('Amplitude')
title('Carrier Signal 3')
ylim([-Cm3 Cm3])
```



```
%% =====
%           CARRIER SIGNALS (FREQUENCY DOMAIN)
%% =====

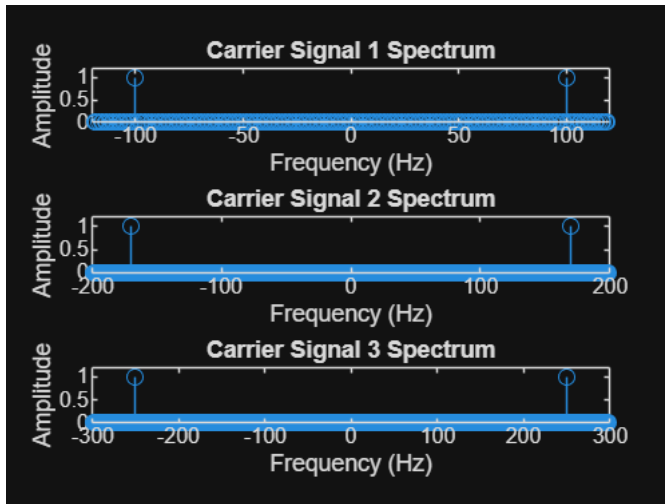
C1 = abs(fftshift(fft(c1)))/(fs/2);
C2 = abs(fftshift(fft(c2)))/(fs/2);
C3 = abs(fftshift(fft(c3)))/(fs/2);

figure;

subplot(3,1,1)
stem(f,C1)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Carrier Signal 1 Spectrum')
axis([-120 120 0 1.2])

subplot(3,1,2)
stem(f,C2)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Carrier Signal 2 Spectrum')
axis([-200 200 0 1.2])

subplot(3,1,3)
stem(f,C3)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Carrier Signal 3 Spectrum')
axis([-300 300 0 1.2])
```

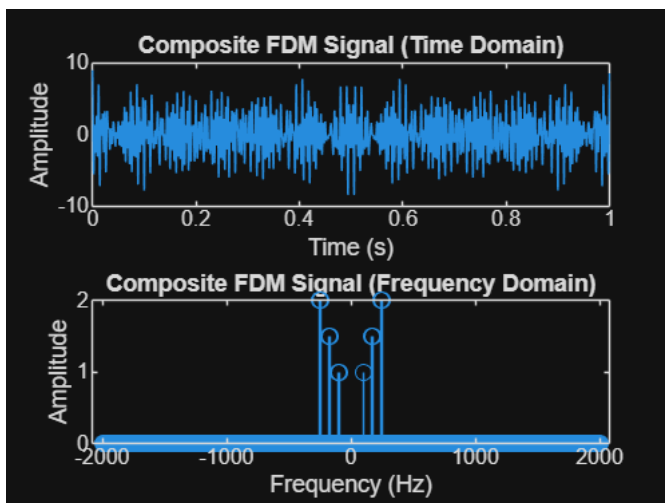


```
%% =====
%                               COMPOSITE FDM SIGNAL
%% =====
```

```
figure;

subplot(2,1,1)
plot(t,x)
xlabel('Time (s)')
ylabel('Amplitude')
title('Composite FDM Signal (Time Domain)')

subplot(2,1,2)
stem(f,X)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Composite FDM Signal (Frequency Domain)')
```



```
%% =====
```

```

%                               BANDPASS FILTERING
%% =====

% Filter for Channel 1
[num1, den1] = butter(5, ...
    [(fc1-fm1-6)/(fs/2), (fc1+fm1+6)/(fs/2)]);
bpf1 = filter(num1, den1, x);

% Filter for Channel 2
[num2, den2] = butter(5, ...
    [(fc2-fm2-6)/(fs/2), (fc2+fm2+6)/(fs/2)]);
bpf2 = filter(num2, den2, x);

% Filter for Channel 3
[num3, den3] = butter(5, ...
    [(fc3-fm3-6)/(fs/2), (fc3+fm3+6)/(fs/2)]);
bpf3 = filter(num3, den3, x);

%% =====
%                               DEMODULATION (MIXING)
%% =====

z1 = 2 * bpf1 .* c1;
z2 = 2 * bpf2 .* c2;
z3 = 2 * bpf3 .* c3;

%% =====
%                               LOW PASS FILTERING
%% =====

[num4, den4] = butter(5, (fm1+3)/(fs/2));
rec1 = filter(num4, den4, z1);

[num5, den5] = butter(5, (fm2+3)/(fs/2));
rec2 = filter(num5, den5, z2);

[num6, den6] = butter(5, (fm3+3)/(fs/2));
rec3 = filter(num6, den6, z3);

%% =====
%                               RECOVERED SIGNALS (TIME DOMAIN)
%% =====

figure;

subplot(3,1,1)
plot(t,rec1)
xlabel('Time (s)')
ylabel('Amplitude')
title('Recovered Signal 1')

```

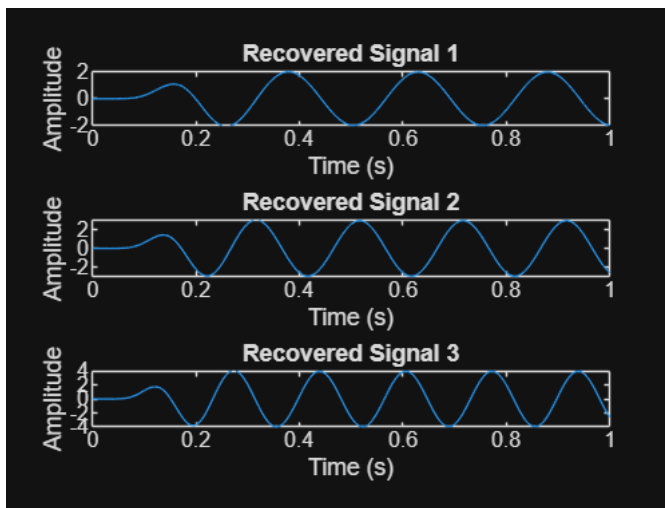
```

ylim([-Am1 Am1])

subplot(3,1,2)
plot(t,rec2)
xlabel('Time (s)')
ylabel('Amplitude')
title('Recovered Signal 2')
ylim([-Am2 Am2])

subplot(3,1,3)
plot(t,rec3)
xlabel('Time (s)')
ylabel('Amplitude')
title('Recovered Signal 3')
ylim([-Am3 Am3])

```



```

%% =====
%               RECOVERED SIGNALS (FREQUENCY DOMAIN)
%% =====

R1 = abs(fftshift(fft(rec1)))/(fs/2);
R2 = abs(fftshift(fft(rec2)))/(fs/2);
R3 = abs(fftshift(fft(rec3)))/(fs/2);

figure;

subplot(3,1,1)
stem(f,R1)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Recovered Signal 1 Spectrum')
xlim([-10 10])

subplot(3,1,2)
stem(f,R2)

```

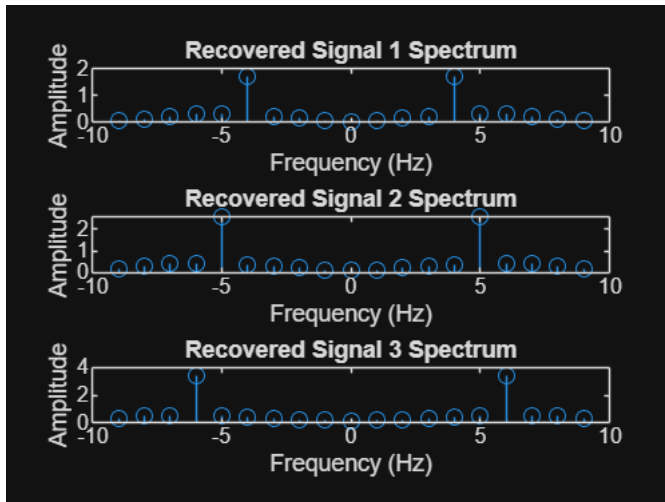


```

xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Recovered Signal 2 Spectrum')
xlim([-10 10])

subplot(3,1,3)
stem(f,R3)
xlabel('Frequency (Hz)')
ylabel('Amplitude')
title('Recovered Signal 3 Spectrum')
xlim([-10 10])

```



```

%% =====
%                                     END OF PROGRAM
%% =====

```